



EE/CE 112: Electric Circuits-I

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Homework 1

Assigned: September 9th, 2024

Due by: September 16th, 2024

Total marks: 40

Important Notes

There are problems for which pen and paper solutions have to be submitted for grading. Preceding the solutions, you must include the following information:

- Your name
- Student ID
- Course ID
- Homework number
- Date and time submitted

Solution should be elaborative, sketch neat circuits wherever required, show clearly all steps of calculation & mention reference (for example formula, law, rule) to strengthen your understanding to solve the electric circuit problems.

Also, please comply with Academic Integrity Policies of Habib University related to plagiarism while submitting this assignment for grading.

Late Submission Policy

You need to submit your solution by the due date. Every day the assignment is late, 10% will be reduced from the total score. Assignment will not be graded after fifth day of submission deadline.

Course Learning Outcomes targeted using this assessment		
CLO	Description	Learning-Domain Level
CLO 1	Identify the fundamentals of basic electrical quantities, circuit terminology, basic circuit laws, and circuit elements in direct current (DC) circuits.	Cog - 2

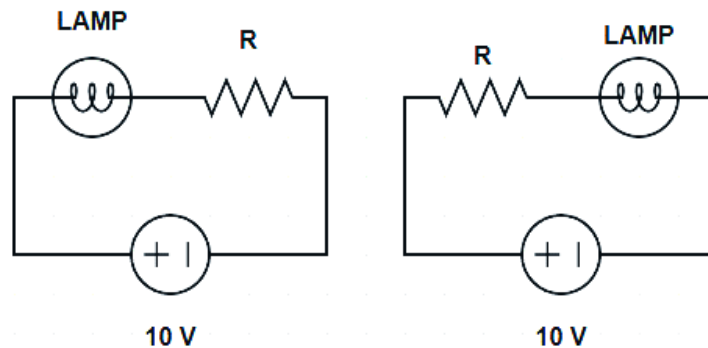


NOTE: Some questions are randomized through your HU ID. Let a, b, c, d and e be the 5 digits of your ID. Then consider the following mapping to attempt questions:

1. $w = b$
2. $x = b + c$
3. $y = b + c + d$
4. $z = b + c + d + e$

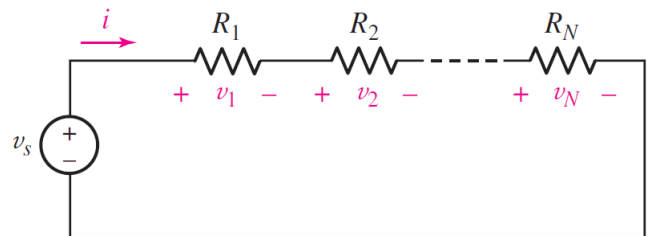
As an example, for HU ID 01234, $w = 1, x = 3, y = 6$ and $z = 10$.

Question 1 [3 pts]: Explain whether there a difference between the two circuits shown below.



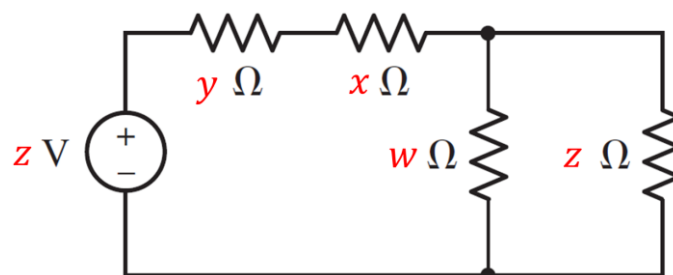
Question 2 [4 pts]: Up till now, we understand that resistors connected in series share the same current, which also means that total voltage is also divided among them. It so happens that we don't have to deal with the current at all when analyzing voltage drops in such circuits, and can directly apply the **Voltage Divider Rule (VDR)** to obtain our results. In general:

$$V_1 = V_S \left(\frac{R_1}{R_1 + R_2 + \dots + R_N} \right)$$



Remember that VDR can only be applied to elements in *series*. Also, elements connected in series form a loop, but not every loop constitutes a series combination.

For the circuit below, use VDR to calculate the individual voltage drop of all resistors.



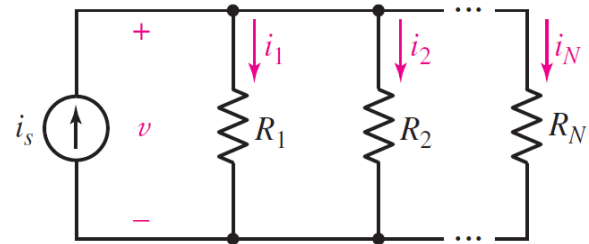
**Question 3 [4 pts]**

When resistors are connected in parallel, as shown in the figure, we have a current divider, for which we can apply the **Current Divider Rule (CDR)**. In general:

$$I_1 = I_s \left(\frac{\frac{1}{R_1}}{\frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_N}} \right)$$

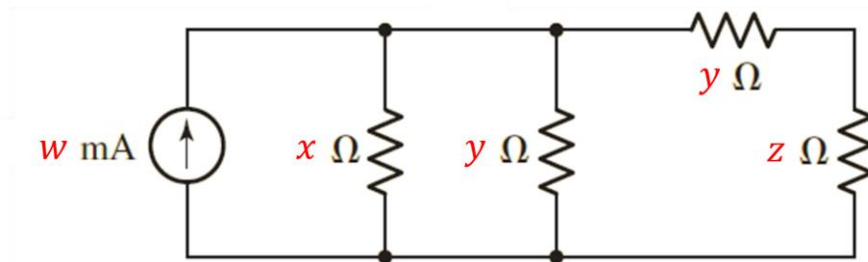
For just two resistors, we can use:

$$I_1 = I_s \left(\frac{R_2}{R_1 + R_2} \right)$$

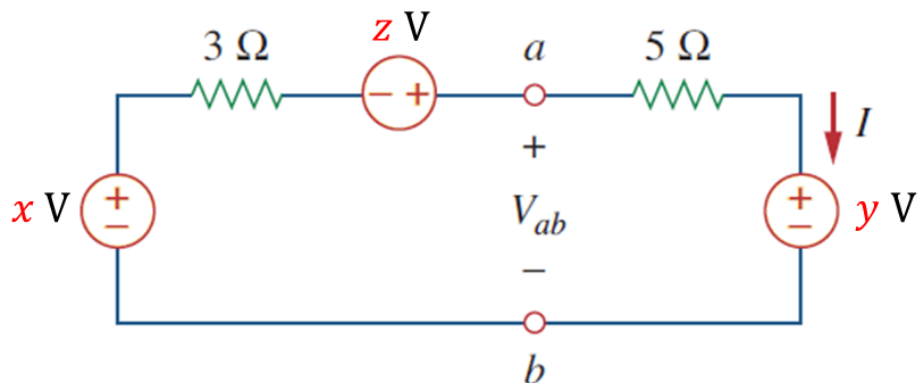


Remember that CDR can only be applied to elements in *parallel*.

For the circuit below, use CDR to calculate all branch currents.

**Question 4 [7 points]:** For the circuit shown below:

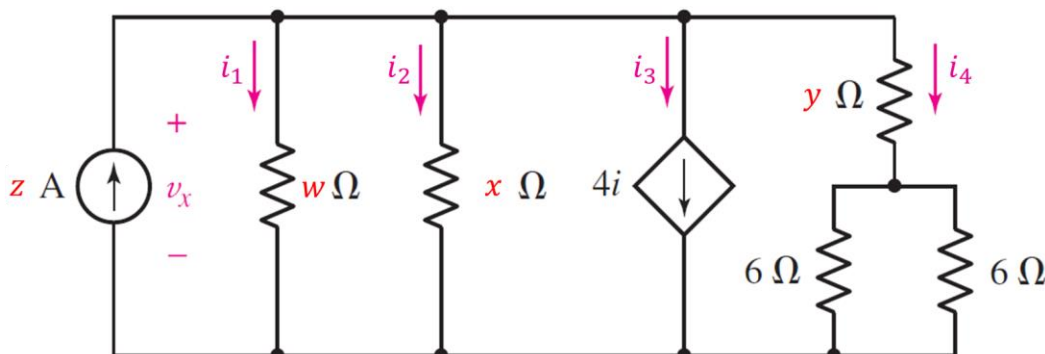
- [3] Identify all nodes, branches and loops.
- [3] Use KVL to find out current I and all unknown voltages.
- [1] What is the value of the voltage V_{ab} ?





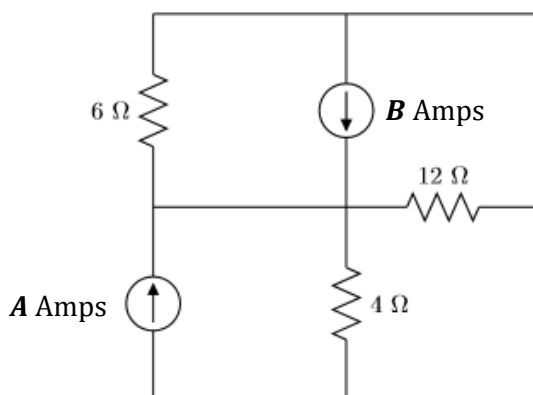
Question 5 [8 points]: For the circuit shown below:

- [3] Identify the number of nodes, loops and branches.
- [4] Use KCL to find out voltage v_x and unknown currents i_1 to i_4 .
- [1] Calculate the power absorbed by the dependent source.



Question 6 [8 points]: For the circuit below, where $A = x + y$ and $B = y + z$:

- [3] How many nodes, branches and loops exist?
- [5] Choose a reference node of your choice, and apply KCL to find the voltage across each element of the circuit.



Question 7 [6 pts]: Three light bulbs are connected to a battery and their power consumption is stated. Calculate:

- [1] The total current supplied by the battery.
- [2] The current through each bulb.
- [3] The resistance of each bulb, assuming that each bulb is treated as a resistor.

